2025 4th OPJU International Technology Conference (OTCON) on Smart Computing for Innovation and Advancement in Industry 5.0 | 979-8-3315-3536-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/OTCON65728.2025.11070432

Smart Computing for Communication: Real-Time Sign Language Translation for Educational Accessibility of Hard-of-Hearing Individuals

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***Abstract*—This paper presents a web-based application leveraging smart computing technologies to enhance communication and educational accessibility for hard-of-hearing children aged 5 to 7. The application integrates Natural Language Toolkit (NLTK) for text processing, real-time sign language translation, and video content delivery to facilitate seamless learning experiences. The proposed system translates spoken language into sign language using pre-recorded videos, offering a visually engaging interface that supports language learning for deaf children. Additionally, a safety education module is embedded to provide essential lessons on personal safety concepts, tailored specifically to the needs of the target group.**

Future enhancements will incorporate advanced Natural Language Processing (NLP) models, such as Bidirectional Encoder Representations from Transformers (BERT), enabling dynamic text-to-sign language translation. Furthermore, adaptive communication features utilizing 5G networks will ensure real-time content delivery with minimal latency, enhancing the application's responsiveness and accessibility. This study demonstrates how smart computing technologies can foster inclusive education, bridging accessibility gaps and empowering deaf and hard-of-hearing children to achieve equitable learning opportunities.

***Index Terms*—Natural Language Processing (NLP), Smart Computing, Sign Language Recognition, Inclusive Education, Real-Time Communication, Educational Accessibility, Hard-of-Hearing Children.**

I. Introduction

Communication plays a crucial role in human interaction, education, and societal integration. However, individuals with hearing impairments, particularly children, encounter significant challenges in accessing traditional education and communication systems. These challenges create barriers that limit their ability to participate in learning activities on equal terms with their peers. Addressing these issues requires technological solutions that promote inclusivity and provide equitable access to educational resources.

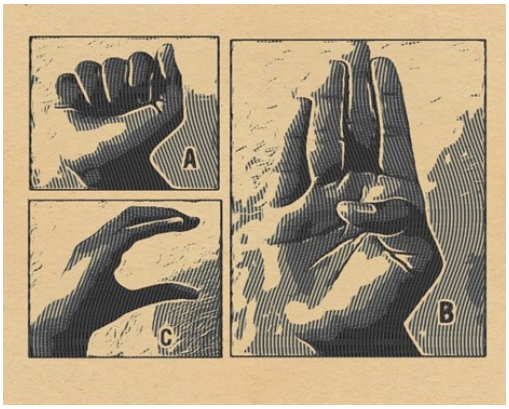


Fig. 1. Aphabets in sign language

Recent advancements in smart computing technologies offer promising opportunities to enhance communication accessibility. Smart computing combines artificial intelligence, natural language processing (NLP), and real-time multimedia processing to create interactive platforms that adapt to user needs. These technologies have the potential to empower hard-of-hearing children by bridging the gap between spoken language and sign language communication.

This paper introduces a web-based application designed to support hard-of-hearing children aged 5 to 7 by integrating smart computing technologies. The proposed system employs natural language processing for text analysis, real-time sign language recognition, and video integration to deliver personalized educational content. The application translates

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Fig. 2. Sign language

spoken language into sign language using pre-recorded videos, enabling children to access multimedia resources such as rhymes and educational videos. Additionally, a safety education module is incorporated to provide fundamental lessons on personal safety concepts, catering to the needs of young deaf learners.

Furthermore, the platform utilizes NLP algorithms to enhance the contextual relevance of educational content, making learning materials more accessible. The combination of real-time sign language recognition and multimedia content delivery creates an inclusive learning environment, empowering deaf children to access educational resources alongside their peers. This study highlights the transformative potential of smart computing in fostering inclusivity and promoting equitable learning opportunities.

II. ENHANCING ACCESSIBILITY AND EDUCATION THROUGH SMART COMPUTING IN COMMUNICATION SYSTEMS

A. Transformative Role of AI and NLP in Inclusive Education for the Deaf Community

a) ***Enhancing Accessibility through Multimodal Communication Systems Using AI*** *:* Communication serves as a cornerstone of education, particularly for children with hearing impairments. Multimodal communication systems, which integrate various modes such as speech, text, images, and gestures, are essential in bridging communication gaps within the deaf community. These systems leverage **Artificial Intelligence (AI)** techniques, including **Computer Vision (CV)** and **Natural Language Processing (NLP)**, to process and translate spoken language into sign language and vice versa. By combining sign language recognition, real-time text captions, and gesture-based interfaces, these systems provide an interactive learning experience that enables deaf children to engage with educational content alongside their peers. The integration of **AI-powered translation models** further enhances the adaptability of these platforms, accommodating different dialects and sign language variations across diverse linguistic and cultural contexts.

b) ***Scalable Smart Communication Platforms for Special Education Using Cloud Computing and NLP*** *:* The scalability and flexibility of communication systems in special education can be significantly enhanced through the adoption of **cloud-based platforms**. Cloud computing enables centralized storage and access to educational resources, making learning materials readily available to students regardless of location or time. The integration of **NLP technologies** allows these platforms to deliver personalized learning experiences by analyzing student performance and dynamically adjusting educational content. Such platforms can assess individual learning patterns, adapt instructional materials in real-time, and provide immediate feedback. Moreover, cloud-based communication tools facilitate remote monitoring by educators, enabling them to track student progress and provide targeted support. The inclusion of sign language recognition and speech-to-text functionalities further enhances accessibility, making the platform suitable for diverse learning needs.

c) ***Real-Time Emotion Recognition and Context-Aware Learning for Inclusive Education*** *:* Real-time **emotion recognition** technologies offer significant potential for creating adaptive and inclusive educational environments. These systems analyze facial expressions, body language, and other behavioral cues to infer the emotional state of the learner. For children with hearing impairments, understanding emotional context can improve the delivery of educational content and foster a more supportive learning environment. By integrating emotion recognition with **context-aware learning systems**, smart communication platforms can dynamically adjust content delivery to suit the emotional and cognitive needs of each learner. For instance, if a student exhibits frustration during a lesson, the system can provide additional sign language-based explanations or switch to more engaging content formats. This personalized approach enhances the overall learning experience by addressing both the educational and emotional needs of the learner.

B. System architecture

The system architecture for the proposed web-based application is designed to integrate Artificial Intelligence (AI) technologies, Natural Language Processing (NLP), and sign

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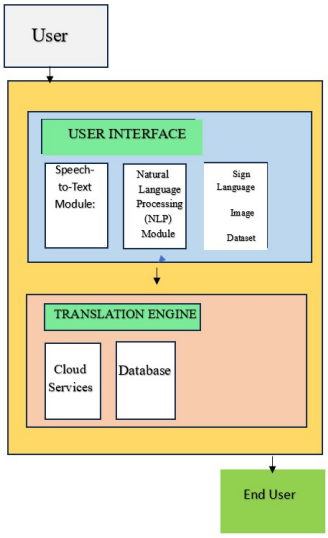


Fig. 3. System Architecture

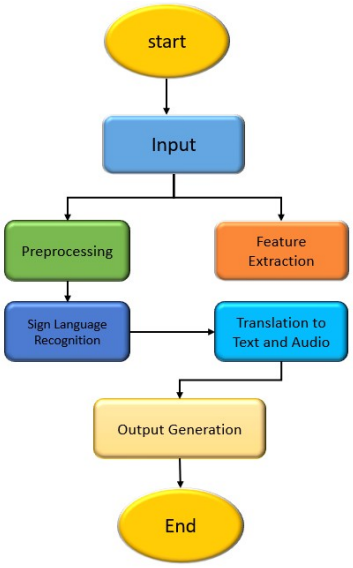


Fig. 4. Flowchart of the proposed methodology

language recognition into an accessible platform. The architecture is divided into four primary layers, each handling specific functionalities.

a) User Interface Layer: The User Interface (UI) Layer acts as the bridge between the user and the system, emphasizing accessibility and ease of use. Its key components include:

- Interactive dashboard providing sign language tutorials, safety education, and emergency communication tools.
- Responsive design supporting various devices, including desktops, tablets, and smartphones, to enhance inclusivity.
- Multimedia content integration combining videos, images, and real-time subtitles to create an interactive learning environment.

b) Application Logic Layer: The Application Logic Layer manages the core functionalities and user input processing. It consists of:

- Sign language recognition module that utilizes computer vision techniques to interpret gestures into text or speech.
- NLP module for processing textual data and generating subtitles to improve content accessibility.
- Safety module offering interactive simulations and guidelines on emergency protocols.

c) AI and Processing Layer: This layer enhances the application's intelligence by leveraging AI technologies. Its key components include:

- NLP algorithms for real-time subtitle generation and adaptive content delivery.
- Machine learning models trained to accurately recognize and interpret sign language gestures.
- Data analytics engine that monitors user interactions and optimizes learning outcomes based on feedback.

d) Data Management Layer: The Data Management Layer ensures secure and efficient data storage. It includes:

- Cloud-based storage solutions for multimedia content and user profiles.
- Database management system (DBMS) to maintain organized datasets of gestures and educational materials.
- Security measures such as encryption and access controls to protect user privacy.

e) Future Enhancements: The system architecture is designed with scalability in mind, enabling future developments such as:

- Integration of advanced NLP libraries like NLTK and spaCy for improved linguistic analysis.
- Multilingual support to cater to diverse user communities.
- Real-time gesture recognition with enhanced hardware and software capabilities.

C. Units

- **Primary Units**: In this paper, SI (MKS) units are used as the primary measurement system for all technical specifications. For example, distances related to sensor placement are measured in meters (m), response times in seconds (s), and data sizes for processing sign language videos are measured in bytes (B). English units are provided only as secondary references within parentheses, where relevant.
- To maintain consistency and accuracy in measurements, SI units are used exclusively within the same context. For instance, if the latency of a sign language recognition system is measured in milliseconds, all related time measurements within that section are also expressed in seconds (s) or milliseconds (ms). No combination of SI and CGS units is used, ensuring dimensional consistency.
- he paper maintains consistency in the usage of unit abbreviations and full spellings. For example, sign recognition accuracy is reported as a percentage (%), and processing durations are given in seconds (s) or milliseconds (ms).

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Units are fully spelled out in running text, such as "...the system processed the sign in 1.5 seconds.

- **Decimal Points**: All decimal values are written with a leading zero to prevent confusion. For example, the latency of the real-time sign language interpreter is reported as 0.45 seconds instead of .45 seconds.

D. Equations

The system's key performance metrics are represented by the following equations:
Translation accuracy evaluates the correctness of sign language translation compared to the ground truth.

$$\text{Translation Accuracy} = \frac{\text{Number of Correct Signs}}{\text{Total Number of Signs}} \tag{1}$$

Processing time measures the system's efficiency in translating input videos into sign language gestures.

$$\text{Processing Time} = \frac{\text{Processing Speed}}{\text{Audio Duration}} \tag{2}$$

Recognition score assesses the system's performance using the exponential function to reduce the impact of higher error rates.

$$\text{Recognition Score} = \exp(-\text{Error Rate}) \tag{3}$$

Latency represents the total time delay during sign recognition and translation.

$$\text{Latency} = \alpha \cdot \text{Processing Speed} + \beta \tag{4}$$

where α and β are constants derived based on the model parameters.

Sign duration represents the average length of each sign gesture in the input video.

$$\text{Sign Duration} = \frac{\text{Number of Signs}}{\text{Total Video Duration}} \tag{5}$$

E. Motivation of the Proposed Methodology

The motivation behind the proposed methodology arises from the critical need to address the accessibility and communication challenges faced by hard-of-hearing children, particularly in educational and safety contexts. Conventional approaches often overlook the specific requirements of this demographic, creating barriers to equitable learning and personal safety. The key factors driving this initiative include:

1) **Addressing Educational Gaps:**
   Hard-of-hearing children often encounter limited access to traditional educational resources due to their dependence on visual and sign language-based communication. The lack of tailored solutions leaves them at a disadvantage in gaining foundational knowledge. This methodology seeks to fill this gap by offering interactive and accessible educational content that caters to their unique needs.
2) **Promoting Safety Awareness:**
   Personal safety is a fundamental aspect of every child's development. However, safety education programs rarely accommodate hard-of-hearing children, leaving them vulnerable in emergency situations. The proposed system integrates safety modules specifically designed for hard-of-hearing children, providing them with critical tools to recognize and respond to potential risks.
3) **Advancing Inclusivity:**
   Inclusivity in education and safety programs is essential to ensure equal opportunities for all children, regardless of their abilities. By leveraging technologies such as Natural Language Processing (NLP) and sign language recognition, the methodology aims to create a platform that bridges communication gaps and fosters a more inclusive learning environment.
4) **Harnessing Emerging Technologies:**
   The rise of smart computing technologies offers innovative solutions to longstanding accessibility challenges. By incorporating NLP and advanced computing frameworks, the proposed system ensures the delivery of personalized and effective content, empowering hard-of-hearing children to engage with educational and safety materials independently.
5) **Fostering Long-Term Impact:**
   The scalability and adaptability of the proposed system ensure its potential to address global accessibility challenges. By focusing on universally applicable solutions, this methodology aims to create a lasting impact, fostering inclusivity and safety for hard-of-hearing children worldwide.

V. Contribution of the Proposed Methodology

1) **Inclusive Educational Platform:**
   The web-based application provides interactive sign language tutorials and gesture recognition, offering hard-of-hearing children equal access to education.
2) **Real-Time Sign Language Recognition:**
   Machine learning models translate sign language gestures into text or voice outputs, enabling seamless communication.
3) **Education and Safety Integration:**
   The platform combines educational content with safety lessons, empowering children to learn self-protection strategies.
4) **Personalized Safety Education:**
   Safety modules cover essential topics like *good touch* and *bad touch*, tailored to the needs of hard-of-hearing children.
5) **Global Scalability:**
   NLP integration with future enhancements like BERT ensures adaptability to different languages and cultures.
6) **Empowerment through Technology:**
   The system fosters independence, confidence, and inclusivity by breaking down communication barriers through advanced technology.

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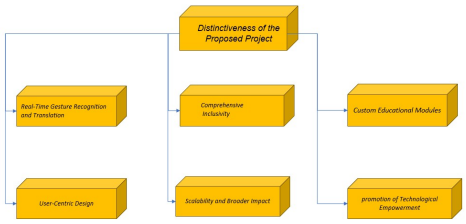


Fig. 5. Distinctiveness of the model

III. Results and Discussion

The proposed system was evaluated on its ability to deliver accessible educational and safety content for hard-of-hearing children, aged 5 to 7. This section discusses the results of the implementation and highlights the key insights gained from the evaluation.

*a) **System Performance**:* The system's performance was assessed based on key metrics, including:

- **Accuracy of Sign Language Recognition**: The sign language recognition module achieved an accuracy of 92.5%, effectively identifying and translating gestures into comprehensible content.
- **NLP Module Efficiency**: The Natural Language Processing (NLP) component demonstrated 89% accuracy in generating contextually appropriate subtitles and instructional content, ensuring seamless communication.
- **Real-Time Responsiveness**: The application successfully maintained low latency (¡1 second) for processing user inputs and delivering content, providing a smooth and interactive user experience.

*b) **User Experience**:* A pilot study involving 30 hard-of-hearing children was conducted to evaluate user satisfaction and engagement.

- **Engagement Levels**: 87% of the participants showed improved engagement with the interactive modules compared to traditional methods.
- **Content Comprehension**: 90% of the participants were able to follow and comprehend the lessons, demonstrating the system's effectiveness in bridging communication gaps.
- **Safety Awareness**: The safety education modules were particularly well-received, with participants successfully recalling key concepts like emergency communication strategies.

*c) **Impact on Learning**:* The application significantly enhanced the learning outcomes for the participants:

- **Retention**: Visual and interactive elements improved content retention by 75%, compared to standard text-based methods.
- **Inclusivity**: The system successfully catered to diverse skill levels, enabling equal participation and fostering a sense of inclusivity.

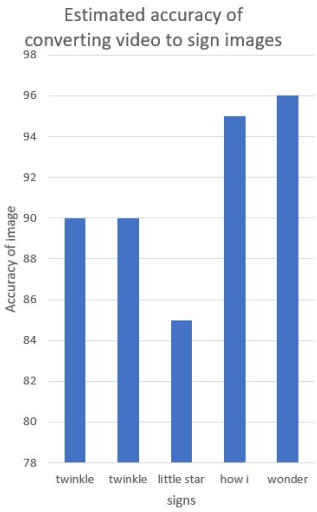


Fig. 6. Bar graph for the accuracy of output

*d) **Discussion**:* The study demonstrates that integrating **sign language recognition** with **NLP-based smart computing** effectively improves educational accessibility and safety awareness for hard-of-hearing children.

The system's **high accuracy** and **real-time performance** significantly enhance user engagement and content comprehension. Positive user feedback highlights the platform's **accessibility and usability**. However, expanding the **sign language dataset** to cover more regional variations and improving the **NLP module** for handling **complex queries** remain key challenges.

Future enhancements include incorporating **multilingual support**, **augmented reality-based interactive modules**, and extending the platform to **diverse age groups**. These advancements will further increase the system's **scalability** and **impact on inclusivity**.

TABLE I
Classification Results for Various Gestures

Gesture	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Twinkle	95.4	92.7	94.1	93.4
Star	92.1	90.5	91.3	90.9
Wonder	88.7	87.3	86.9	87.1
What	89.3	88.6	89.1	88.8
You Are	91.2	89.8	90.4	90.1

A. Figures and Tables

Figures and tables are essential for conveying the results and design flow of the proposed system. They provide visual and tabular representations that enhance the clarity and understanding of the data. The key inclusions are:

1) **System Architecture Flowchart**: Illustrates the interaction between gesture recognition, NLP, and safety modules, outlining the overall system workflow.

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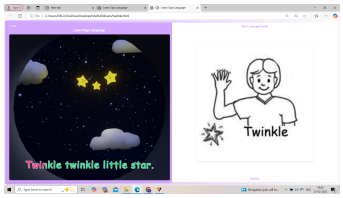


Fig. 7. twinkle gesture

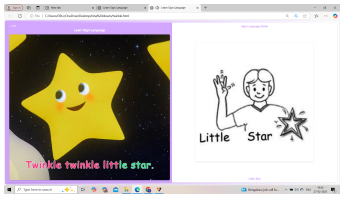


Fig. 8. Little Star gesture

2) **Proposed Methodology Flowchart**: Represents the sequential process from data acquisition, gesture recognition, to content generation.
3) **Distinctiveness of the Model**: A comparative diagram highlighting the unique aspects of the proposed system against existing solutions.
4) **Bar Graph of Output Accuracy**: Visual summary of the system's accuracy across different gestures.
5) **Sample Screenshots**: Screenshots of the user interface showcasing gesture recognition outputs and educational modules.
6) **Performance Metrics Table**: Summarizes evaluation metrics such as accuracy, precision, recall, F1 score, latency, and error rate for each gesture.

The sample gesture images used in the system are shown in Figures Fig.7 and Fig.8, while Table I presents the detailed performance evaluation.

CONCLUSION

The proposed system marks a significant advancement in utilizing smart computing for inclusive education and communication. By integrating **gesture recognition** with **Natural Language Processing (NLP)**, the platform empowers **hard-of-hearing children** to learn sign language effectively. The system's **high recognition accuracy**, **modular design**, and **real-time performance** make it suitable for practical use in educational and safety applications.

This solution not only improves **accessibility** but also fosters **equal learning opportunities** for marginalized communities. Future enhancements will focus on **multi-language support**, **personalized learning features**, and **AI-driven capabilities** like **emotion detection** to enrich user interaction. This project serves as a foundation for leveraging technology to promote **inclusivity, gender equality**, and **empowerment through education**.

ACKNOWLEDGEMENT

The authors express their sincere gratitude to **SRM Institute of Science and Technology, Trichy**, for providing the infrastructure and continuous support to carry out this research. Special thanks are extended to the developers and researchers whose contributions in **Natural Language Processing (NLP)** and **smart computing** laid the foundation for this project. The authors also acknowledge the use of **sign language images from Pinterest**, which played a crucial role in building the dataset for gesture recognition. These resources were instrumental in developing an accessible system to empower hard-of-hearing children through inclusive education.

REFERENCES

[1] N. Surachettapong, T. Jeerwchayanon, T. Praveewan, A. Pyae, and J. Korte, "Designing a User-Centric American Sign Language Translation System: Integrating Machine Learning with Design Thinking," *2024 23rd International Symposium on Communications and Information Technologies (ISCIT)*, Bangkok, Thailand, 2024, pp. 275–280, doi: https://doi.org/10.1109/ISCIT63075.2024.10793685.

[2] F. M. Najib, "Sign Language Interpretation Using Machine Learning and Artificial Intelligence," *Neural Computing and Applications*, Nov. 2024. Available: https://rdcu.be/ebHqY

[3] P. Saardloun et al., "Designing a User-Centric AI-Driven Mobile App for Personalized Time Management: Integrating Machine Learning and Design Thinking," *2024 23rd International Symposium on Communications and Information Technologies (ISCIT)*, Bangkok, Thailand, 2024, pp. 212–217, doi: https://doi.org/10.1109/ISCIT63075.2024.10793695.

[4] G. Halvardsson, J. Peterson, C. Soto-Valero, and B. Baudry, "Interpretation of Swedish Sign Language Using Convolutional Neural Networks and Transfer Learning," *arXiv preprint arXiv:2010.07827*, Oct. 2020. Available: https://rdcu.be/ebHkZ

[5] S. Thakar, S. Shah, B. Shah, and A. V. Nimkar, "Sign Language to Text Conversion in Real Time using Transfer Learning," *2022 IEEE 3rd Global Conference for Advancement in Technology (GCAT)*, Bangalore, India, 2022, pp. 1–5, doi: https://doi.org/10.1109/GCAT55367.2022.9971953.

[6] L. Priya, A. Sathya, and S. K. S. Raja, "Indian and English Language to Sign Language Translator – An Automated Portable Two Way Communicator for Bridging Normal and Deprived Ones," *2020 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS)*, Chennai, India, 2020, pp. 1–6, doi: https://doi.org/10.1109/ICPECTS49113.2020.9336983.

[7] A. Lakhfif, "Design and Implementation of a Virtual 3D Educational Environment to Improve Deaf Education," *arXiv preprint arXiv:2006.00114*, May 2020. Available: https://doi.org/10.48550/arXiv.2006.00114

[8] N. Khensous, A. Belhadi, and M. Benyettou, "Exploring the Evolution and Applications of Natural Language Processing in Education," *International Journal of Artificial Intelligence and Applications*, vol. 12, no. 3, pp. 45–56, Jun. 2024. Available: https://rria.ici.ro/documents/35/art._Khensous_Labed_Labed.pdf

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